

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

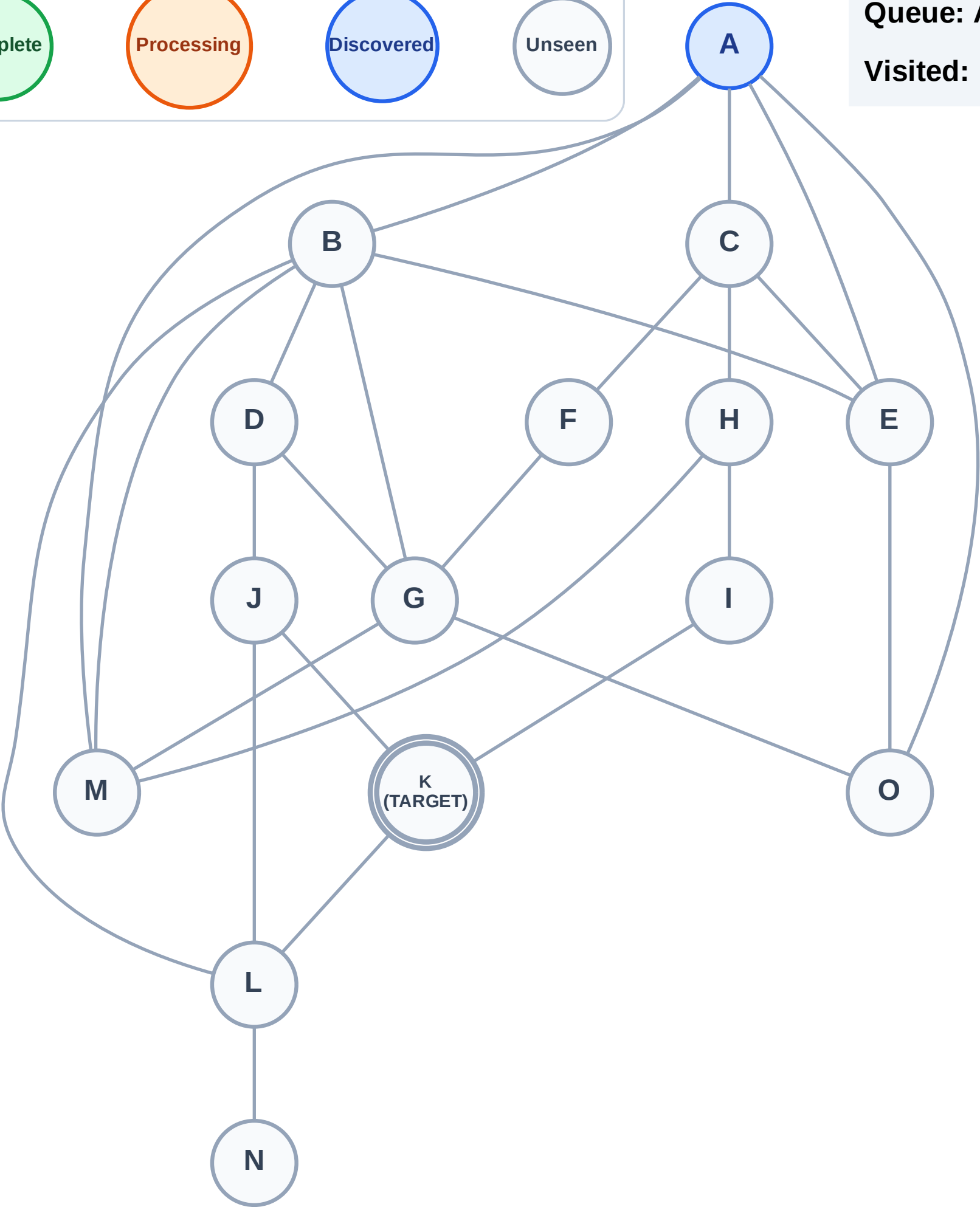
Complete

Processing

Discovered

Unseen

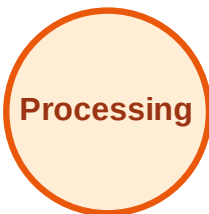
Queue: A
Visited: (none)



BFS Traversal

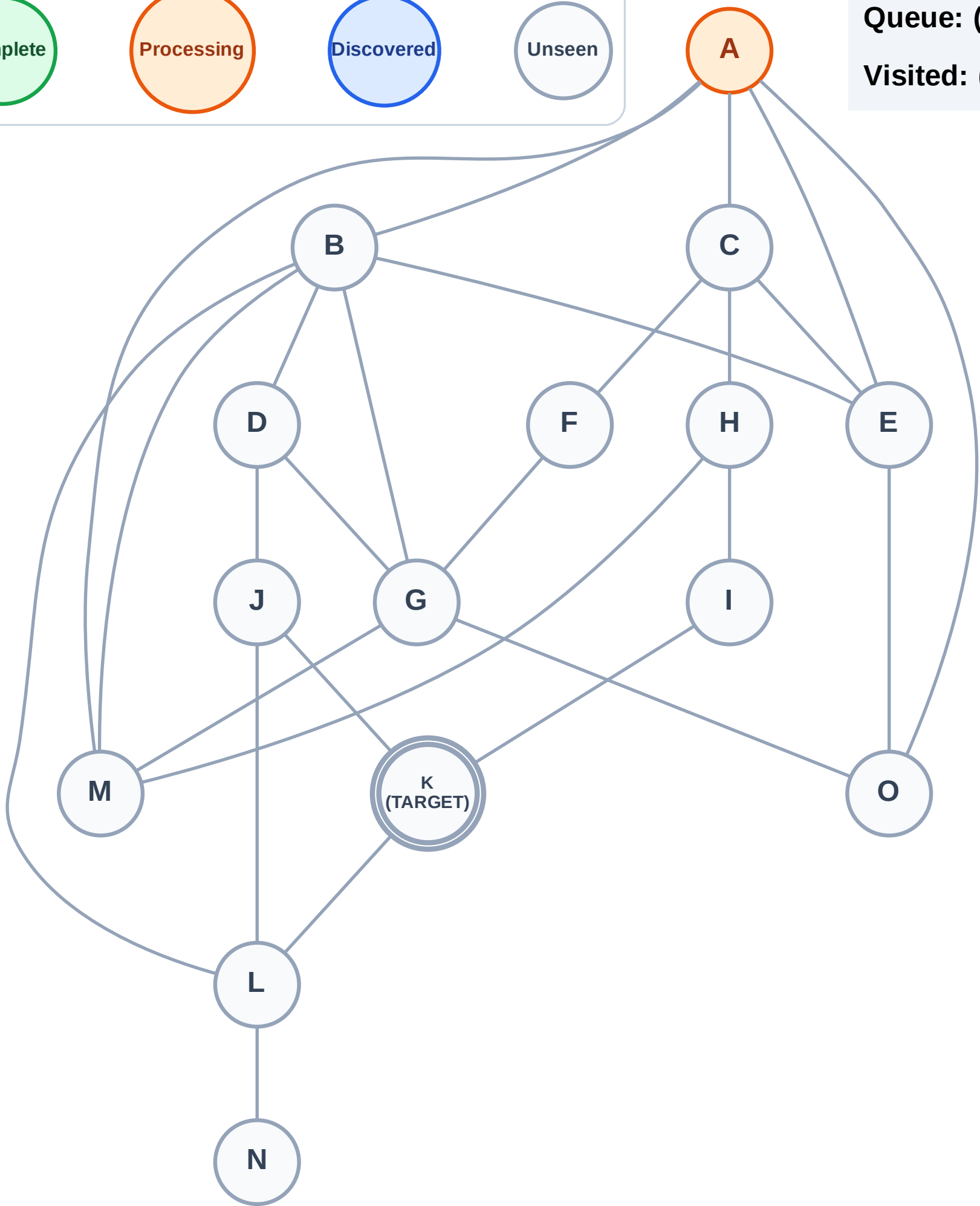
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, M, O from A

Legend

Complete

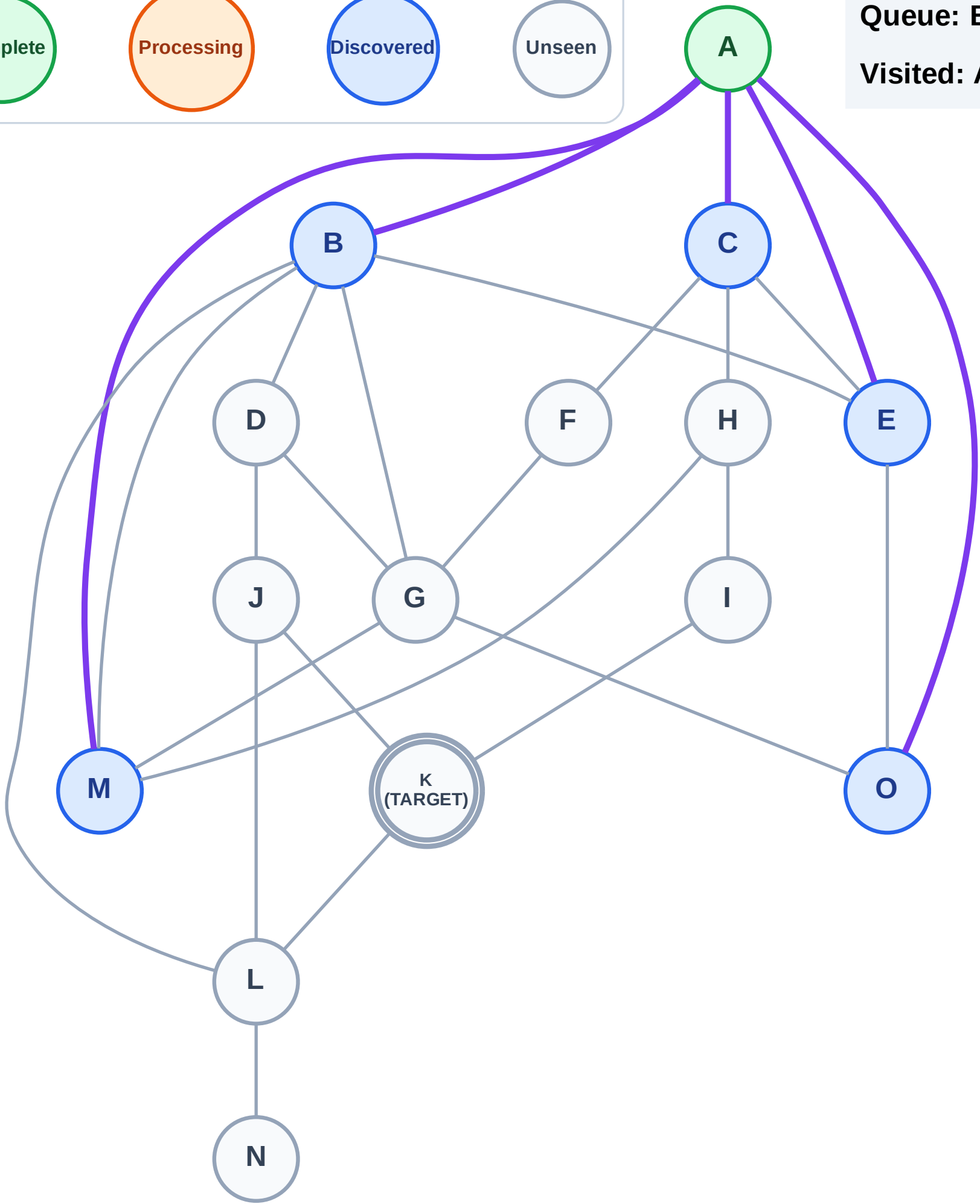
Processing

Discovered

Unseen

Queue: B → C → E → M → O

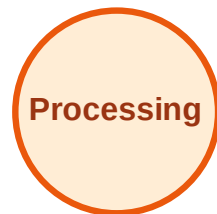
Visited: A



BFS Traversal

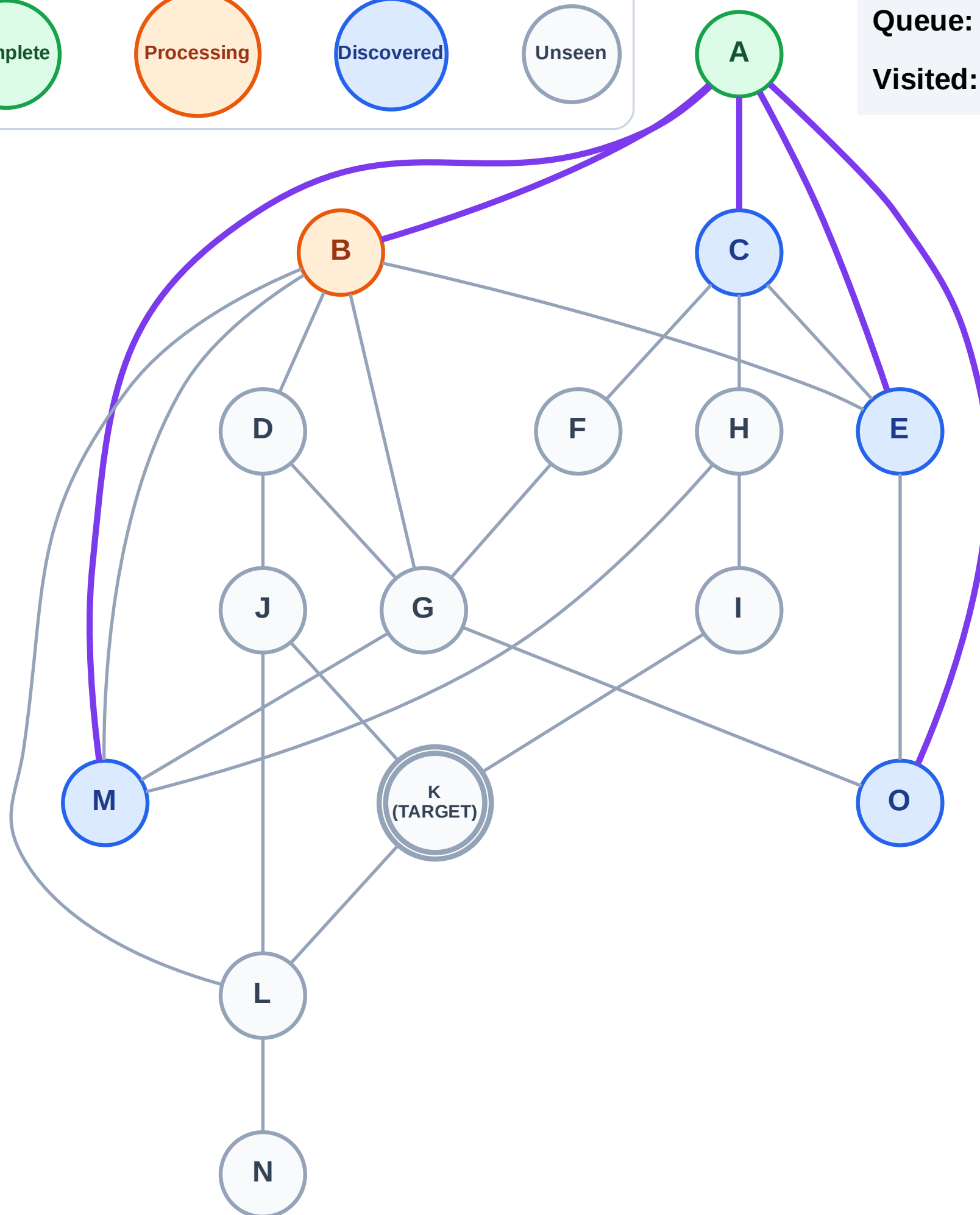
Step 4: Process node B

Legend



Queue: C → E → M → O

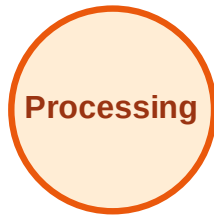
Visited: A



BFS Traversal

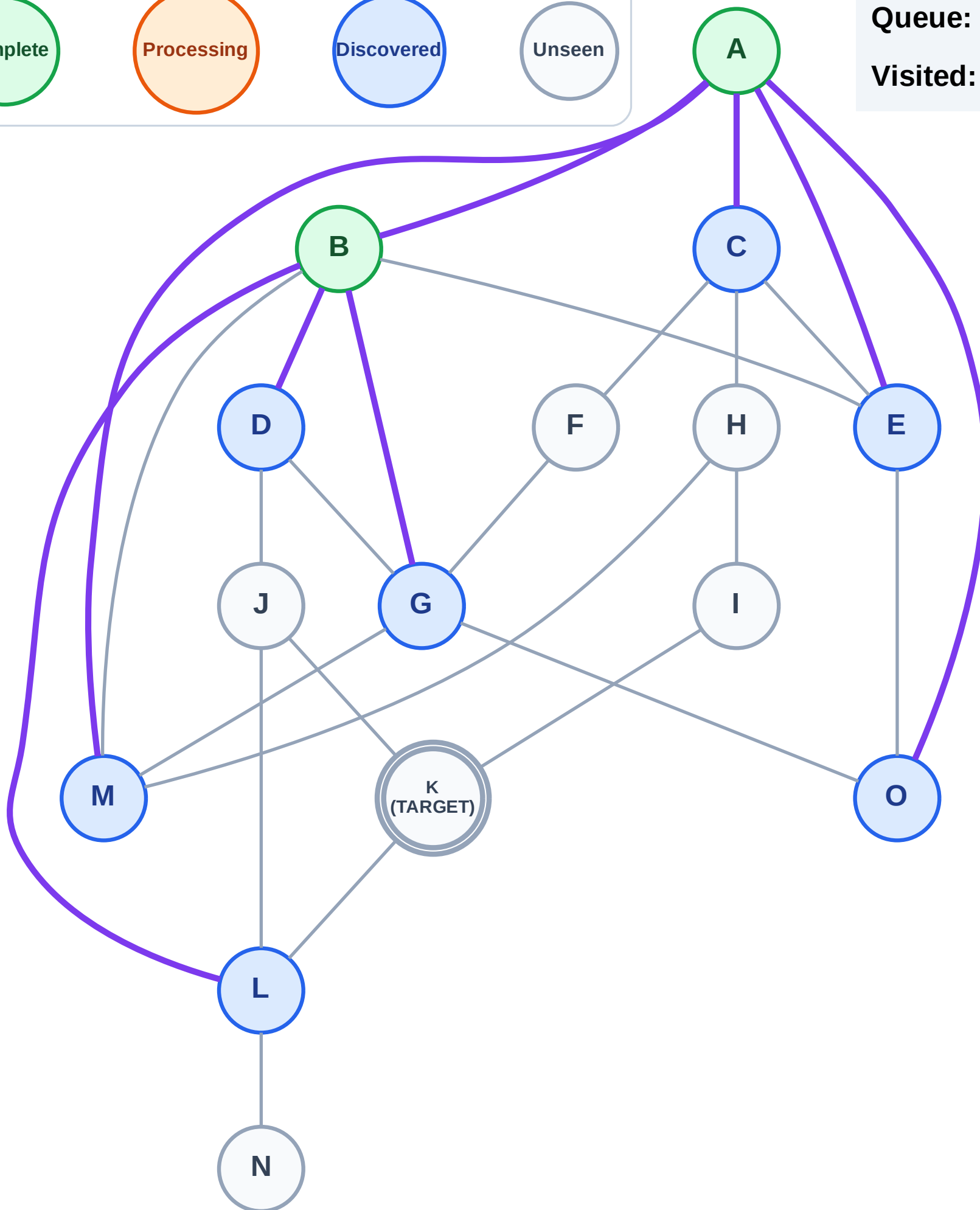
Step 5: Discover D, G, L from B

Legend



Queue: C → E → M → O → D → G → L

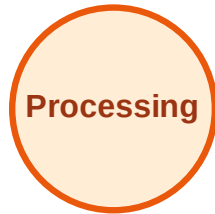
Visited: A, B



BFS Traversal

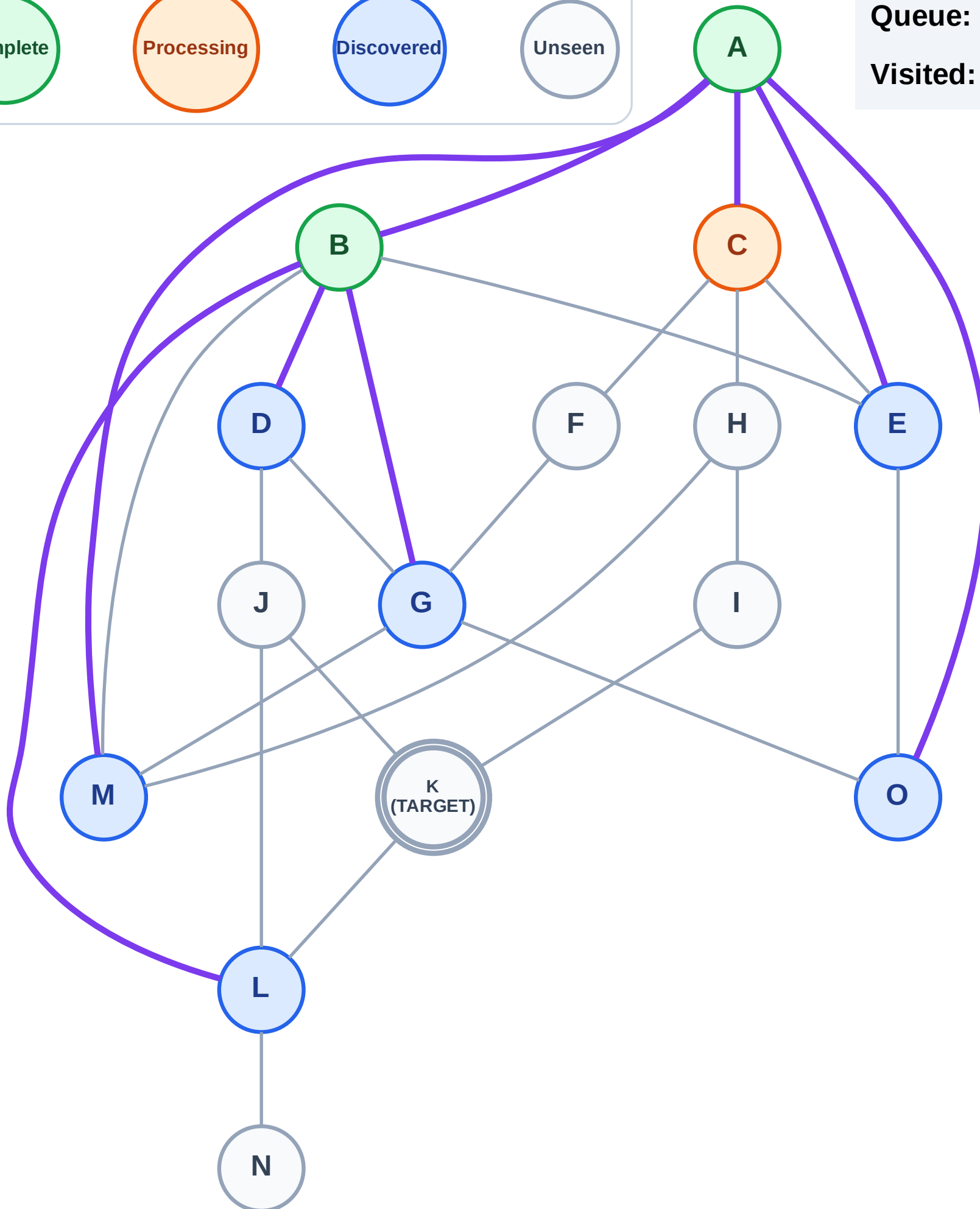
Step 6: Process node C

Legend



Queue: E → M → O → D → G → L

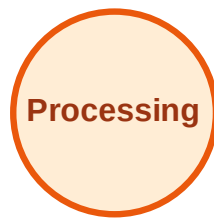
Visited: A, B



BFS Traversal

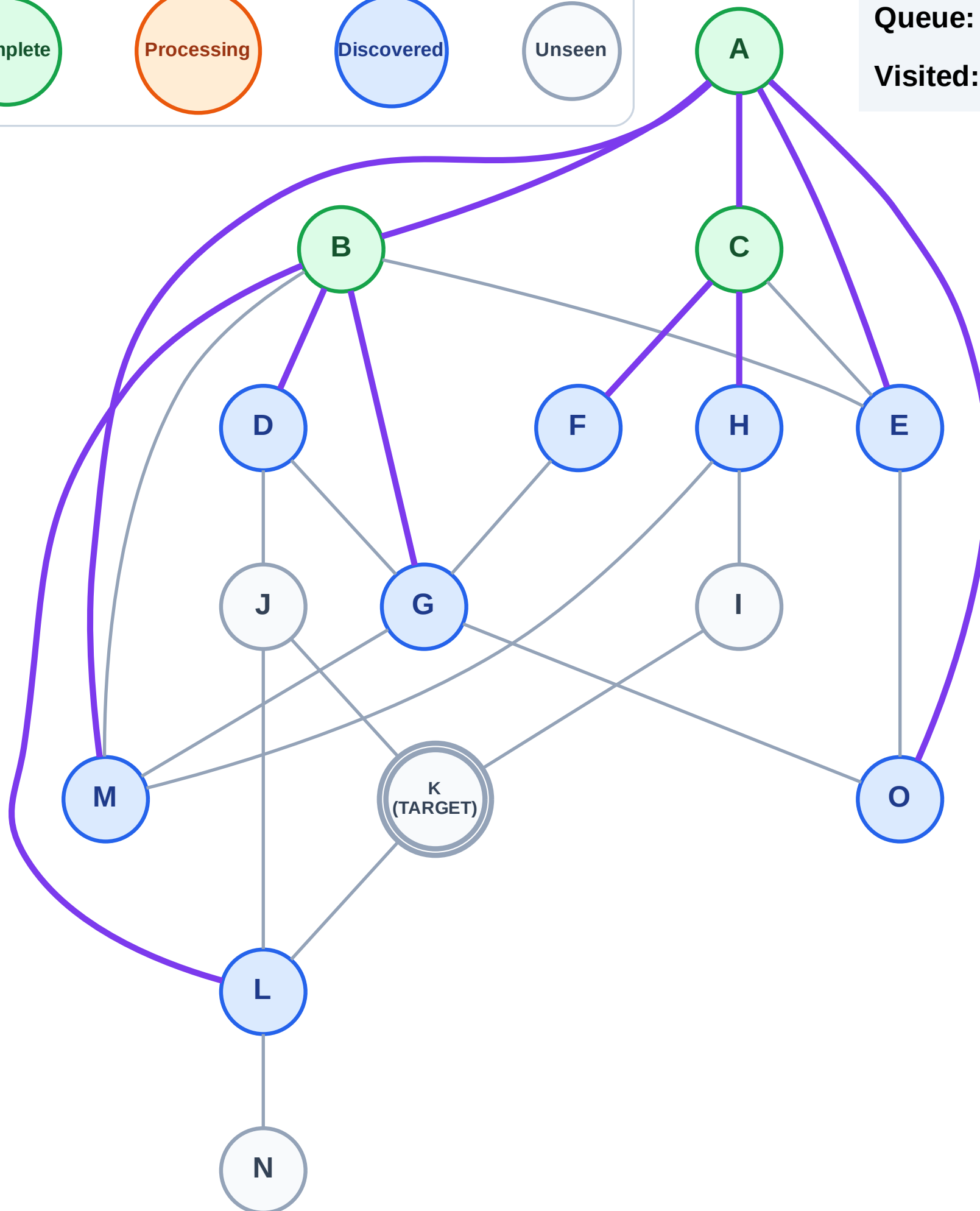
Step 7: Discover F, H from C

Legend



Queue: E → M → O → D → G → L → F → H

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

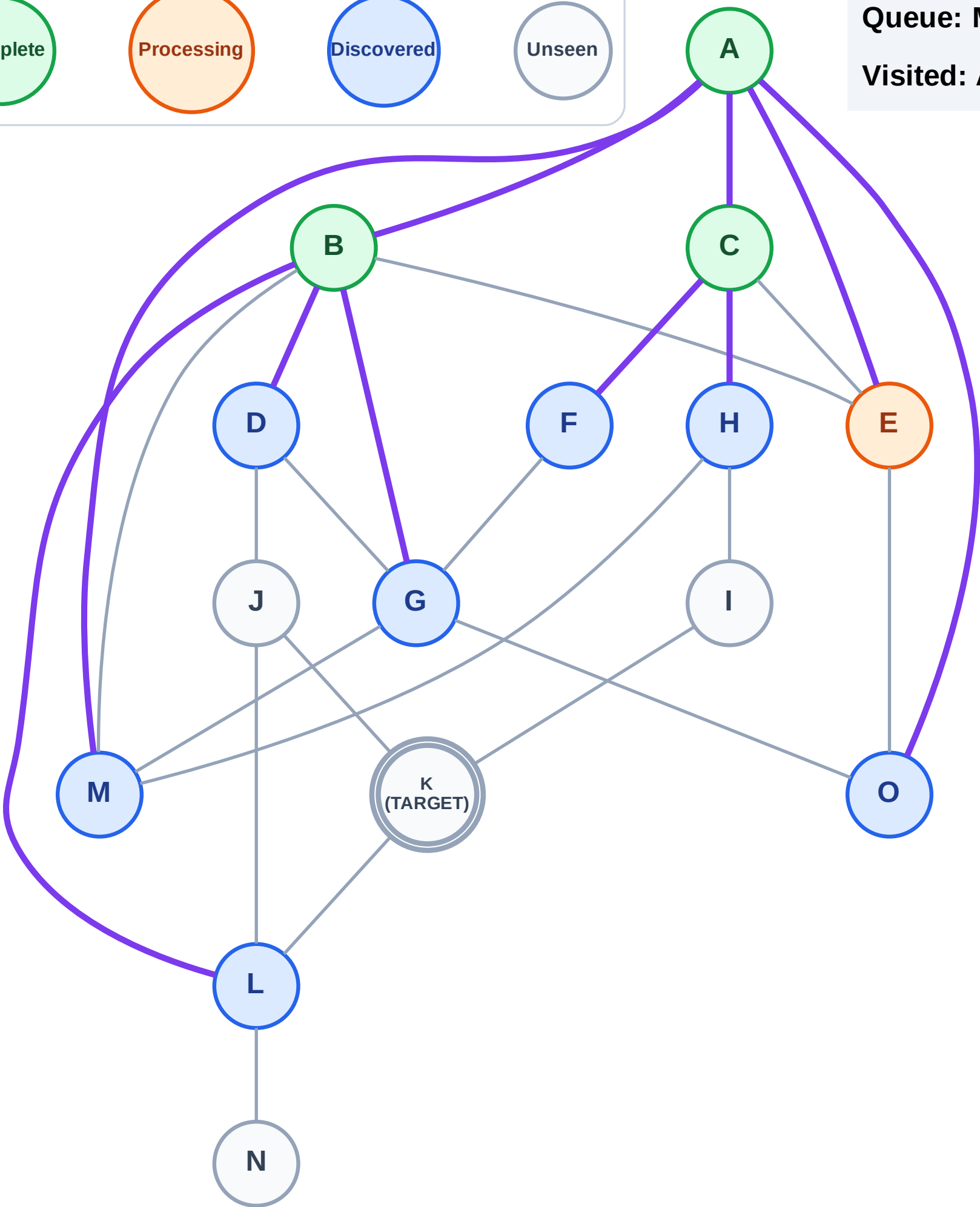
Processing

Discovered

Unseen

Queue: M → O → D → G → L → F → H

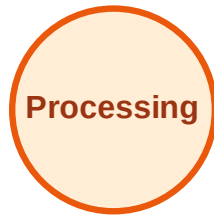
Visited: A, B, C



BFS Traversal

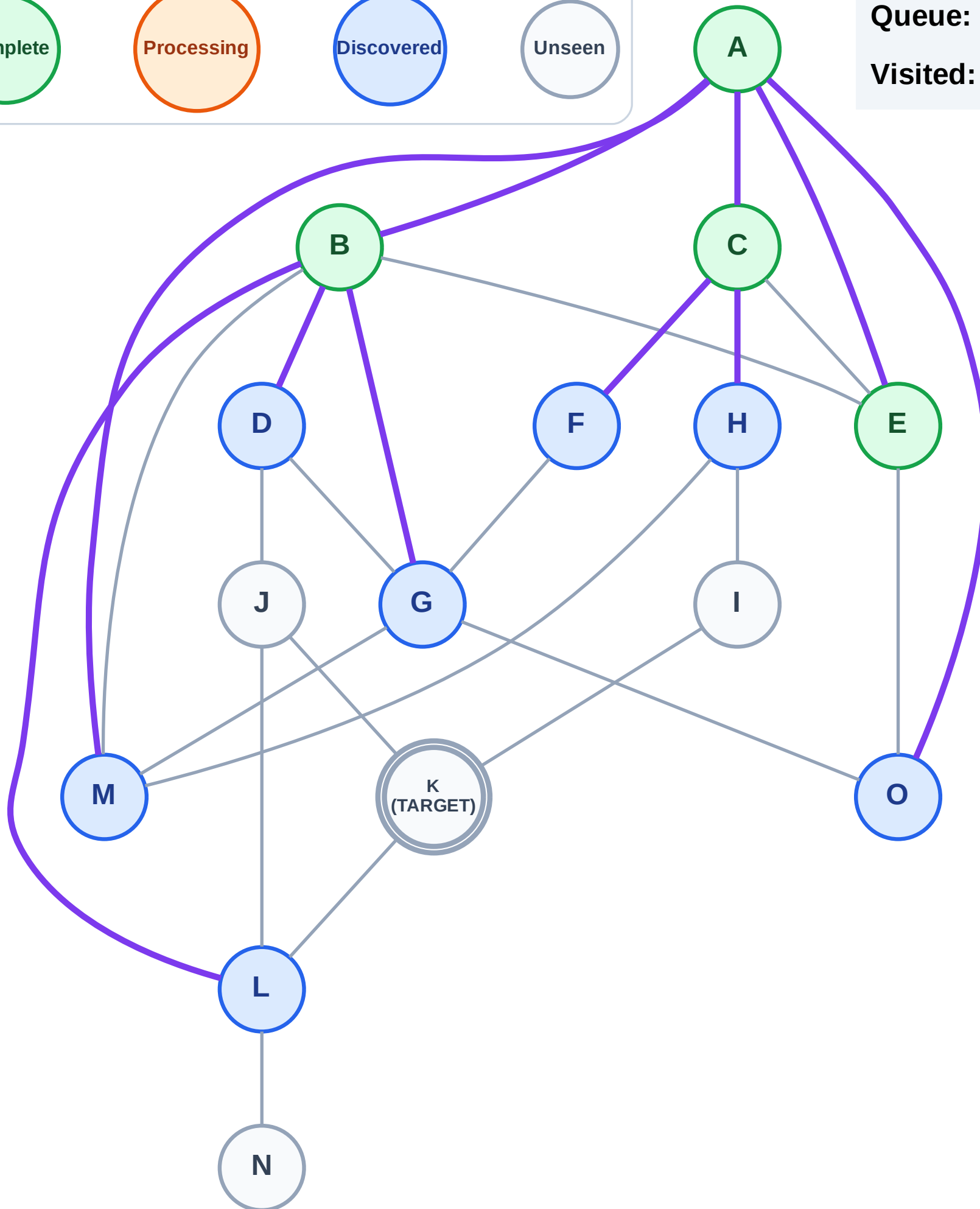
Step 9: No new neighbors from E

Legend



Queue: M → O → D → G → L → F → H

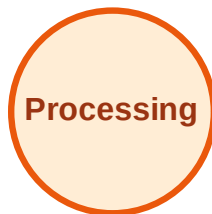
Visited: A, B, C, E



BFS Traversal

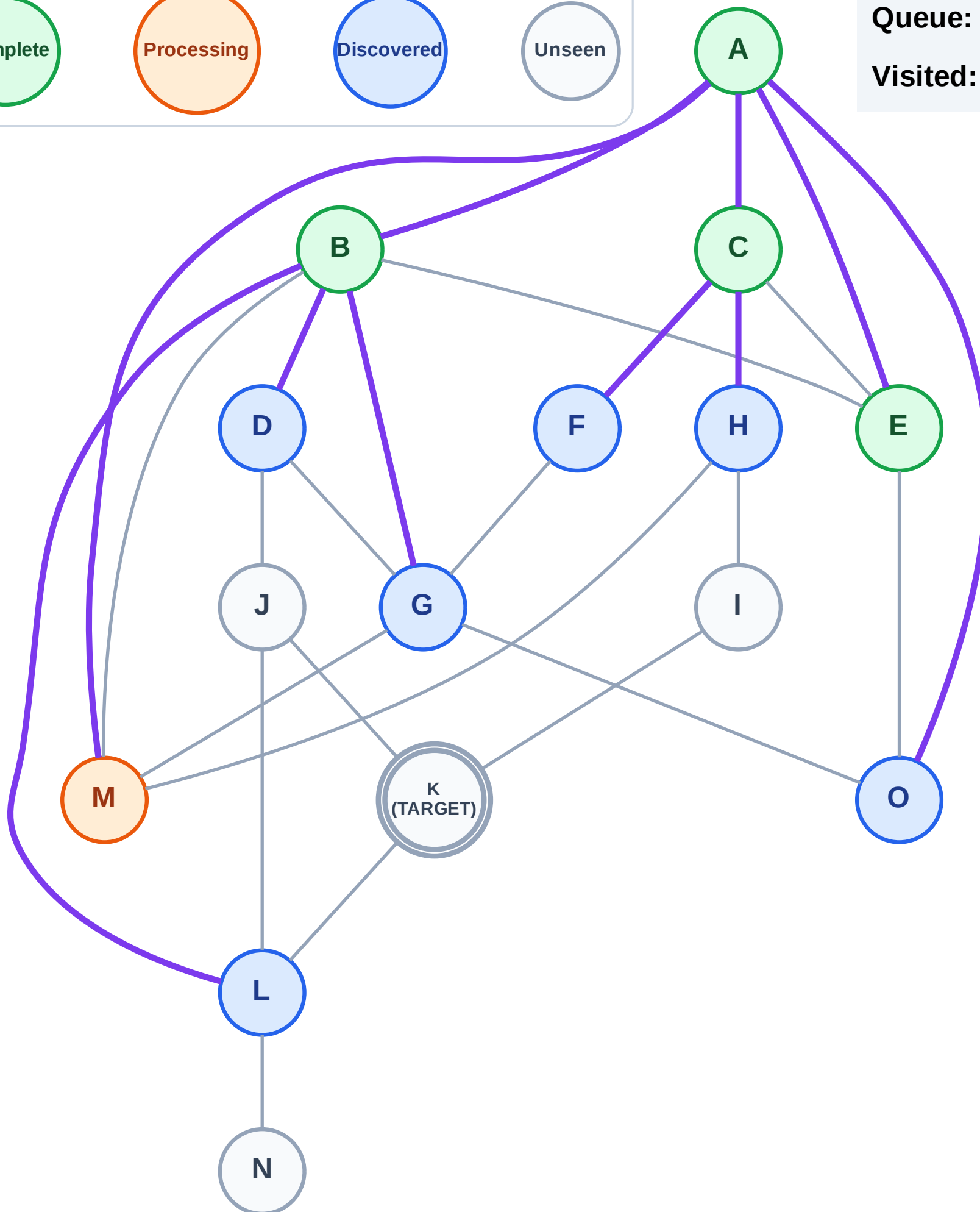
Step 10: Process node M

Legend



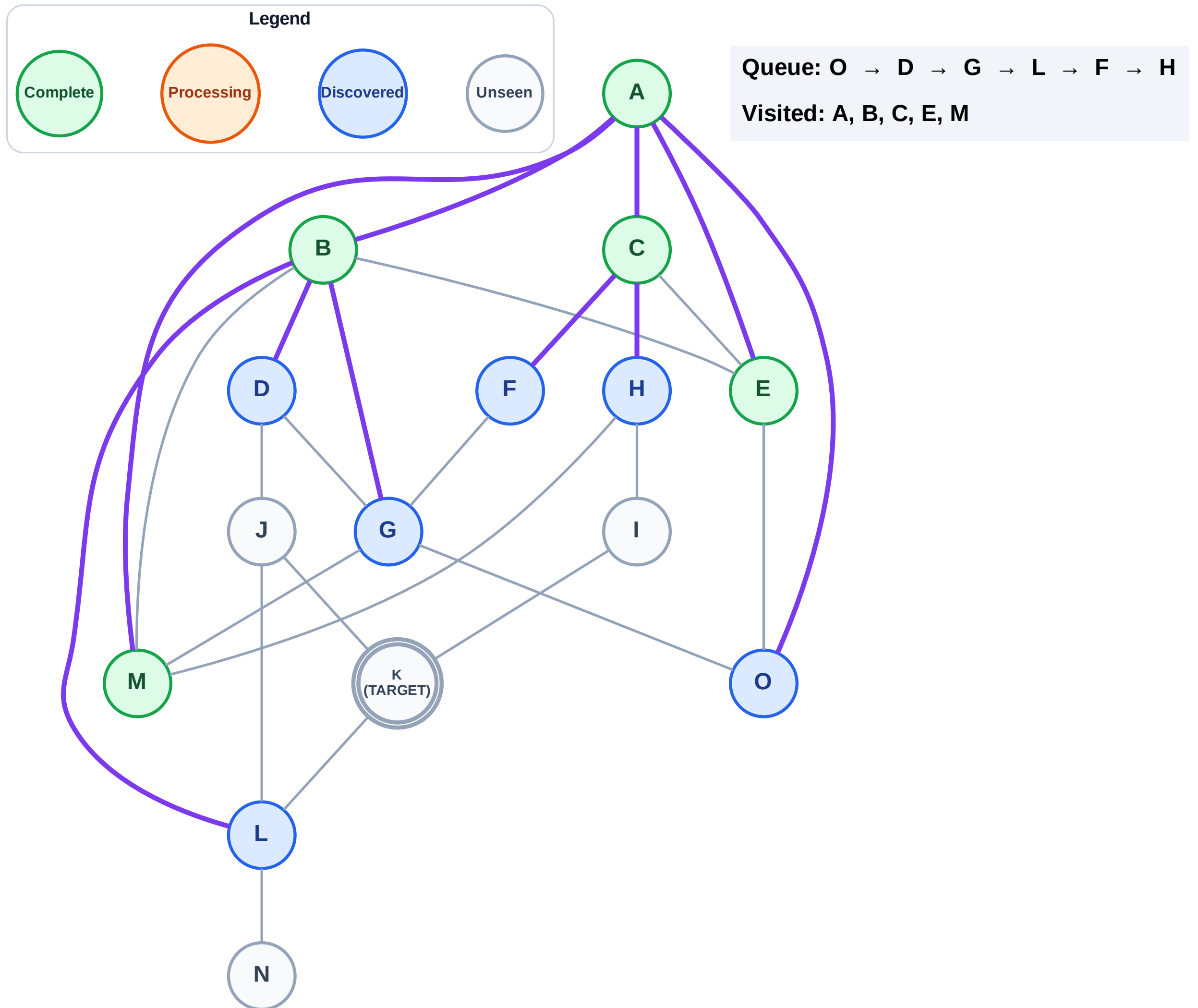
Queue: O → D → G → L → F → H

Visited: A, B, C, E



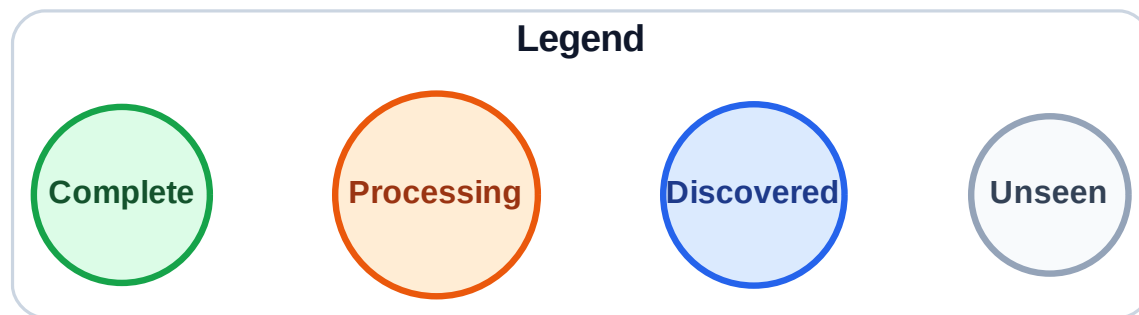
Step 11: No new neighbors from M

Step 11: No new neighbors from M



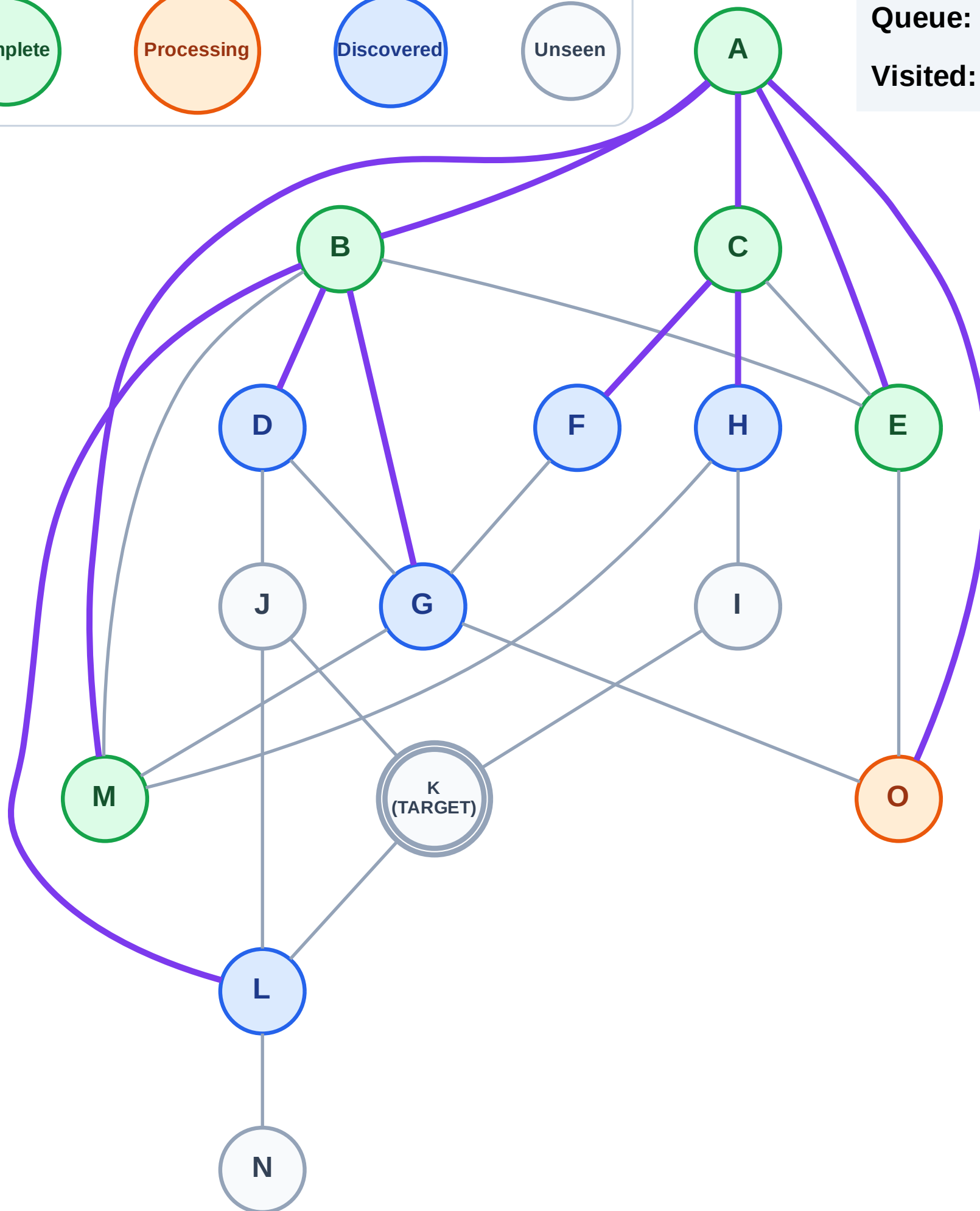
BFS Traversal

Step 12: Process node O



Queue: D → G → L → F → H

Visited: A, B, C, E, M



BFS Traversal

Step 13: No new neighbors from O

Legend

Complete

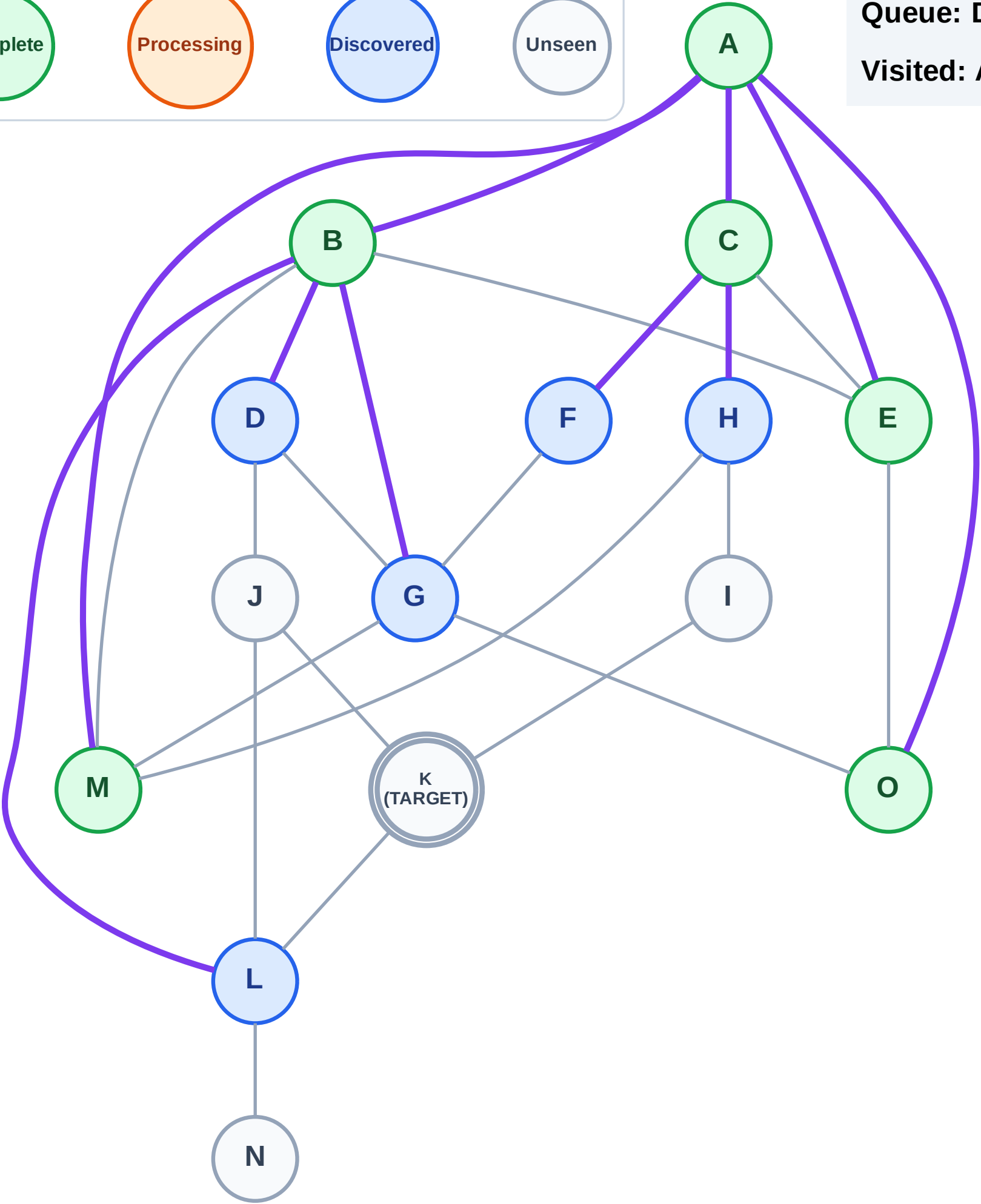
Processing

Discovered

Unseen

Queue: D → G → L → F → H

Visited: A, B, C, E, M, O



BFS Traversal

Step 14: Process node D

Legend

Complete

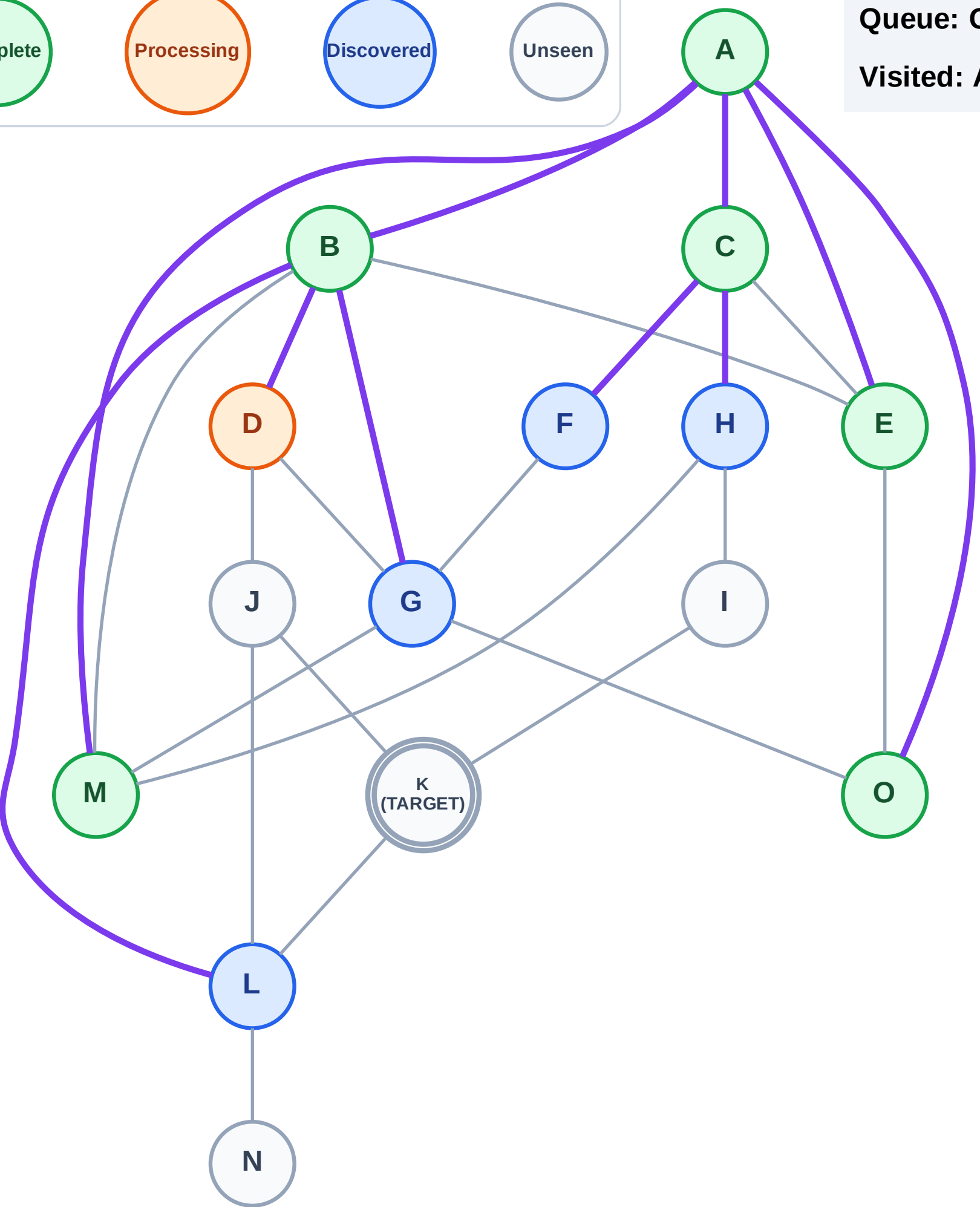
Processing

Discovered

Unseen

Queue: G → L → F → H

Visited: A, B, C, E, M, O



BFS Traversal

Step 15: Discover J from D

Legend

Complete

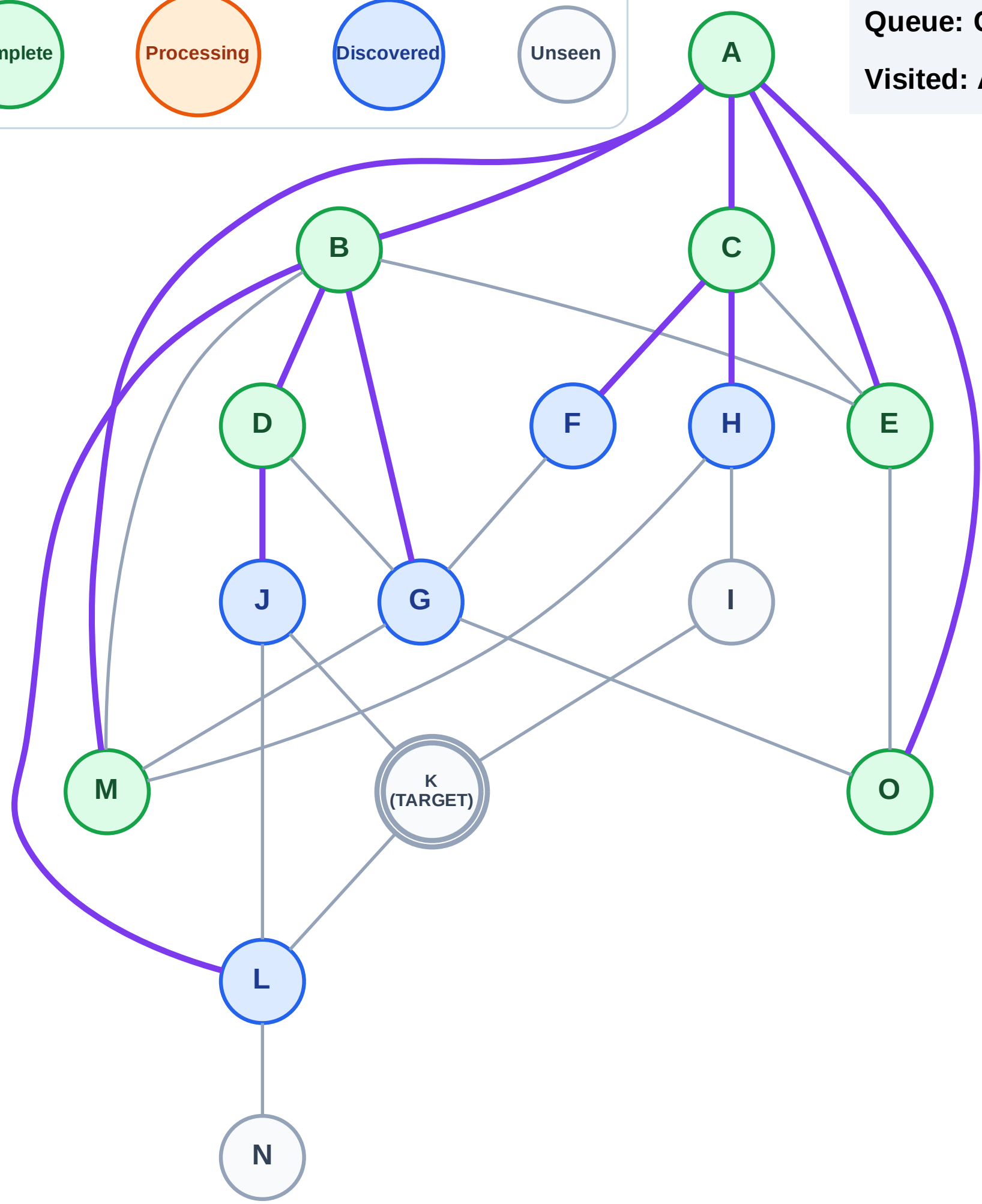
Processing

Discovered

Unseen

Queue: G → L → F → H → J

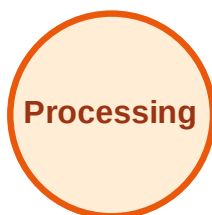
Visited: A, B, C, D, E, M, O



BFS Traversal

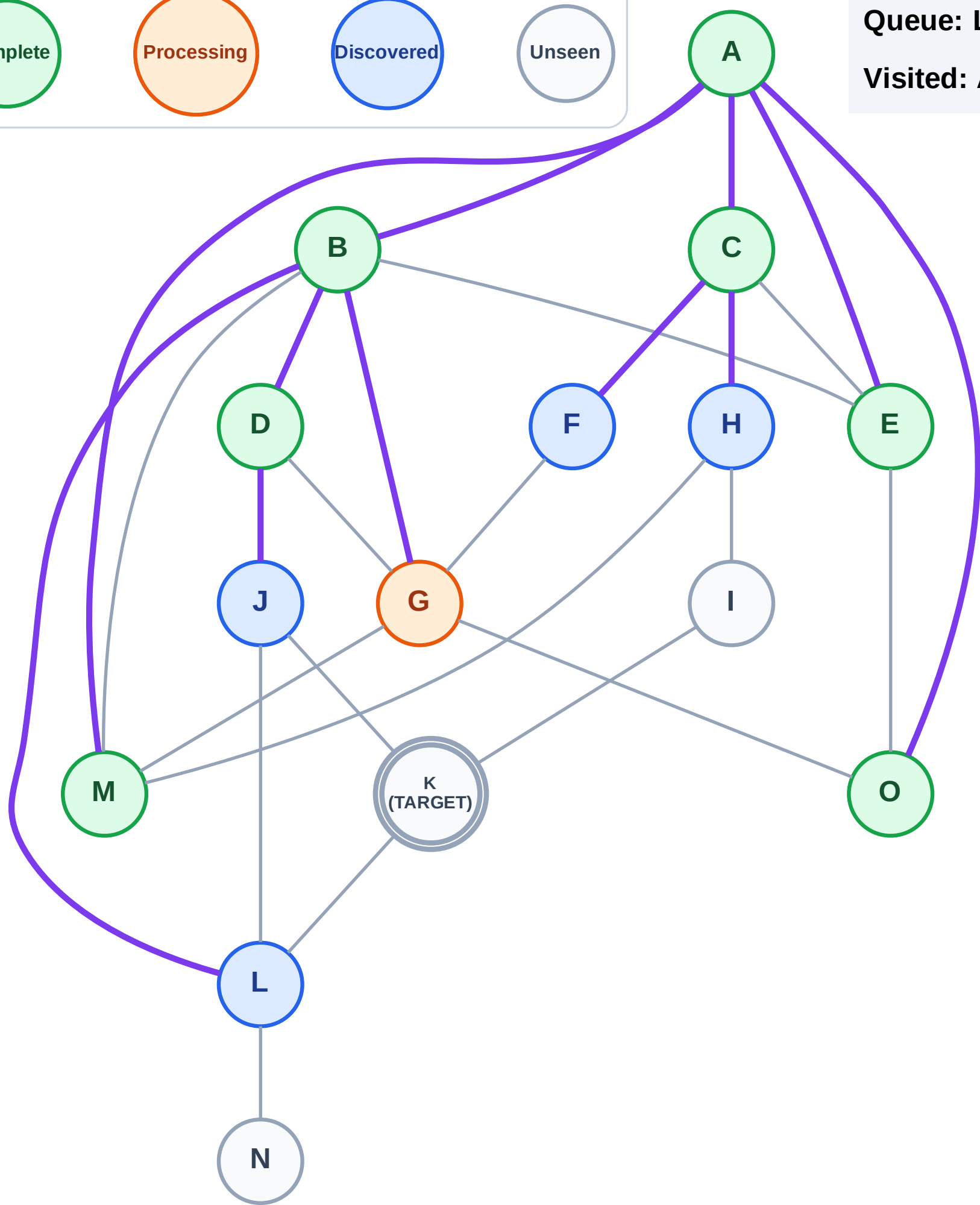
Step 16: Process node G

Legend



Queue: L → F → H → J

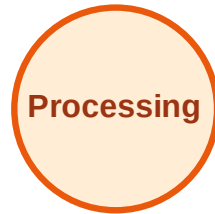
Visited: A, B, C, D, E, M, O



BFS Traversal

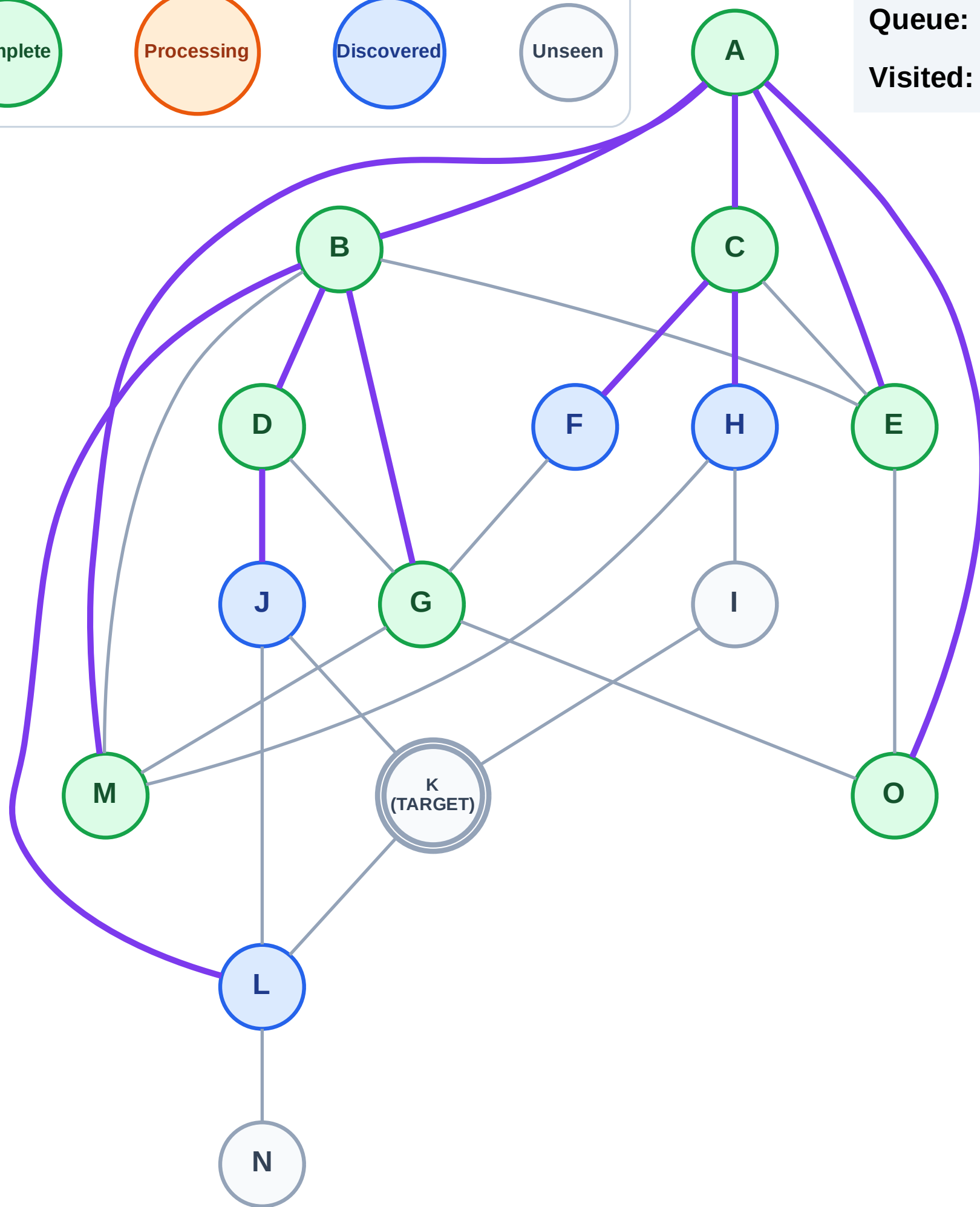
Step 17: No new neighbors from G

Legend



Queue: L → F → H → J

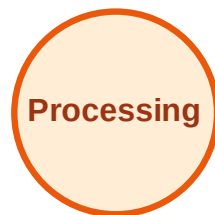
Visited: A, B, C, D, E, G, M, O



BFS Traversal

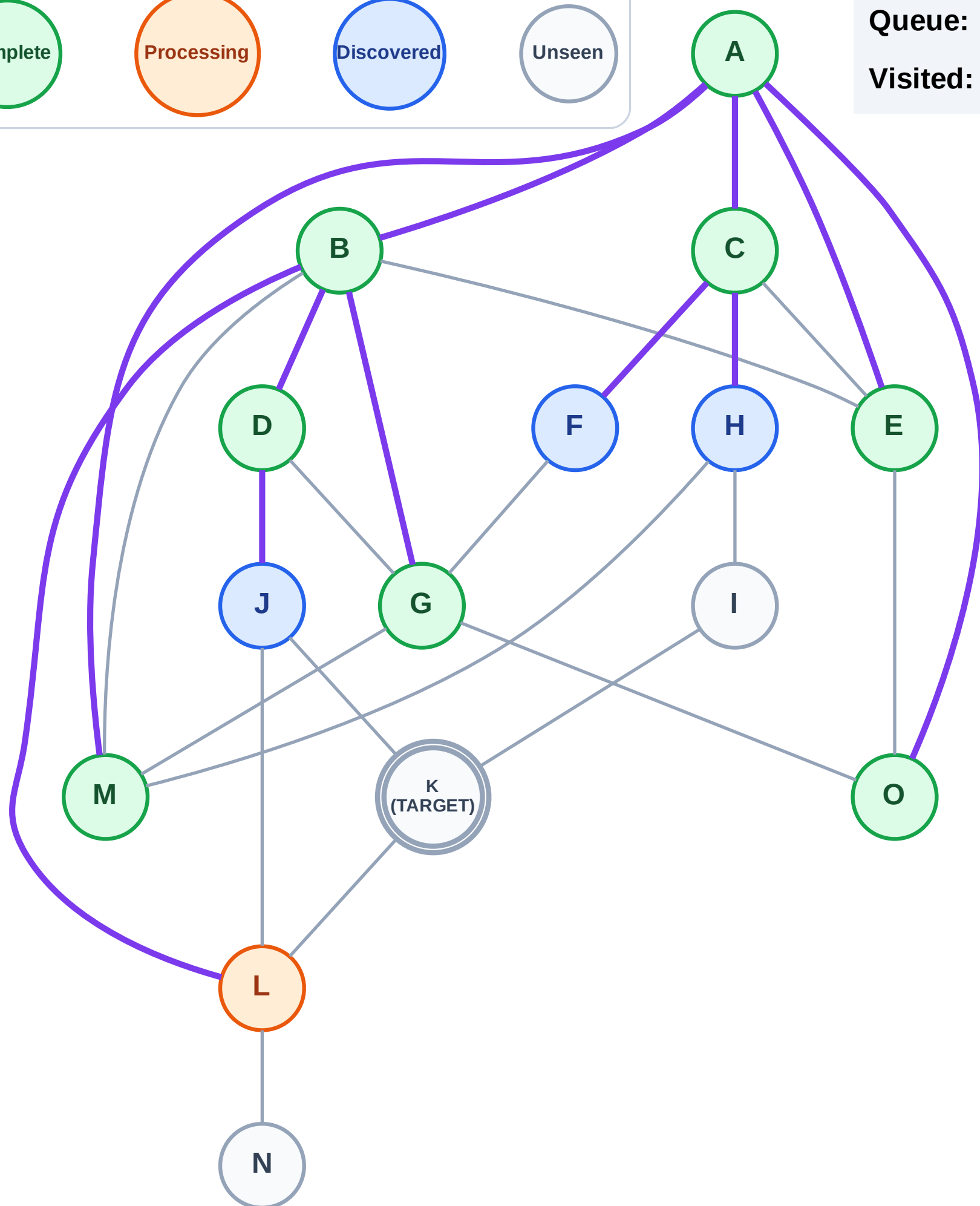
Step 18: Process node L

Legend



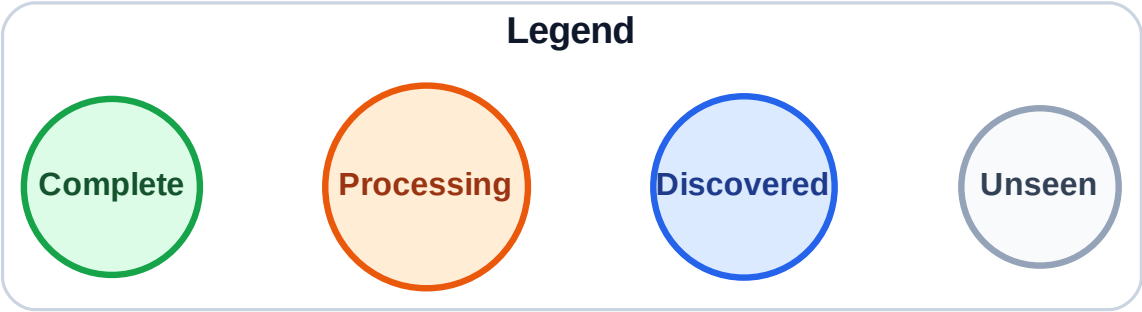
Queue: F → H → J

Visited: A, B, C, D, E, G, M, O

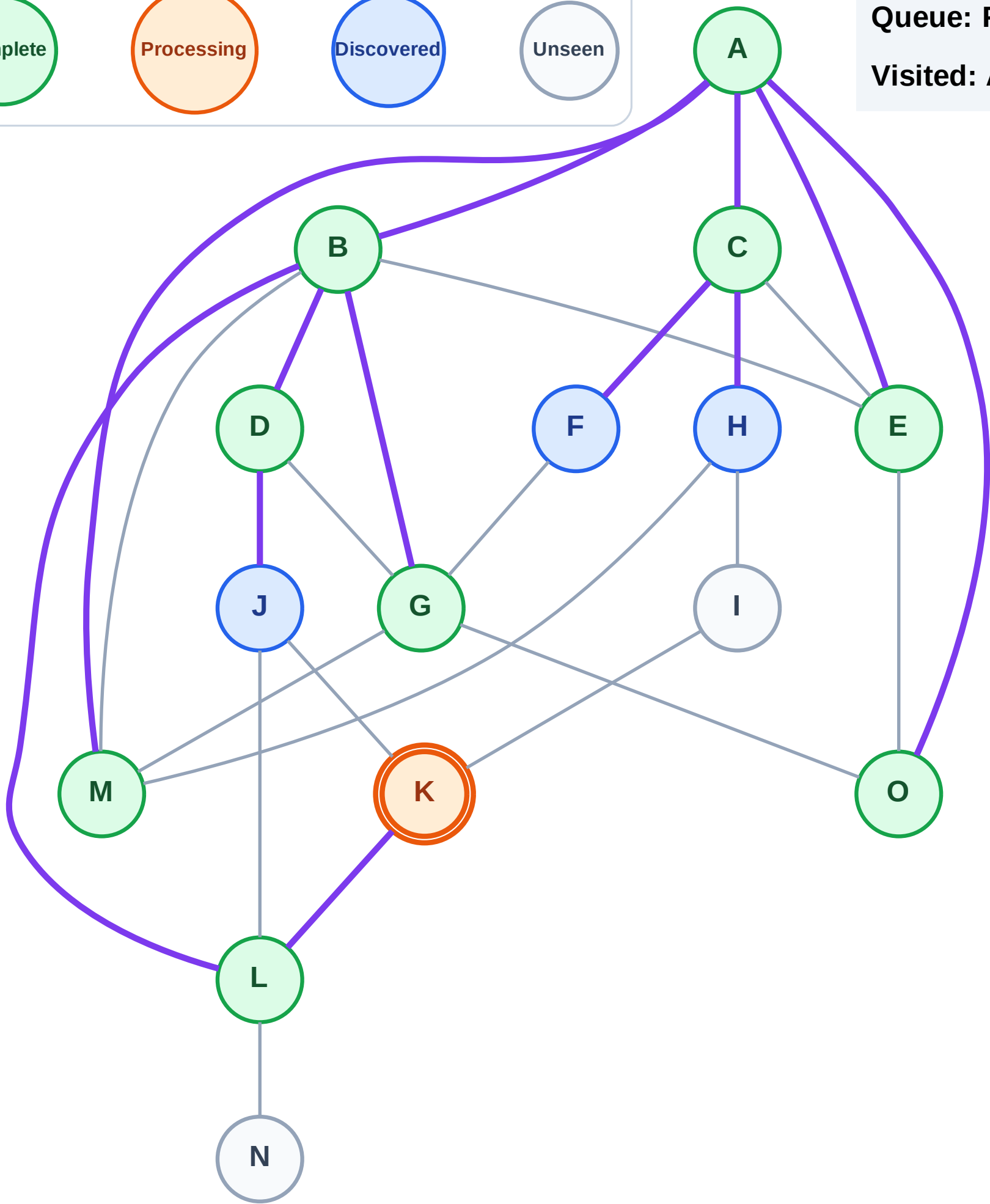


BFS Traversal

Step 19: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → L → K



Queue: F → H → J
Visited: A, B, C, D, E, G, K, L, M, O



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

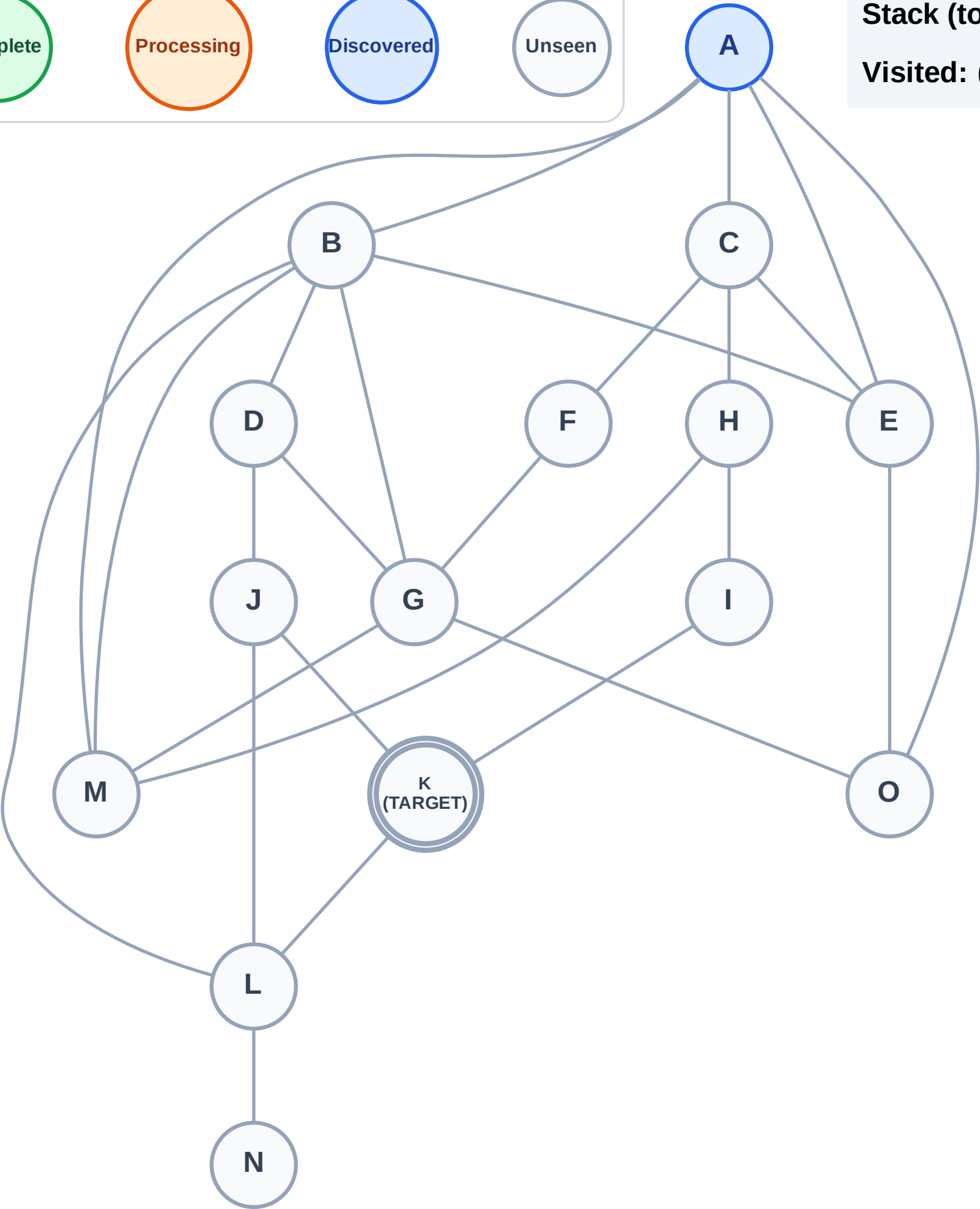
Complete

Processing

Discovered

Unseen

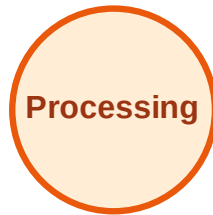
Stack (top at right): A
Visited: (none)



DFS Traversal

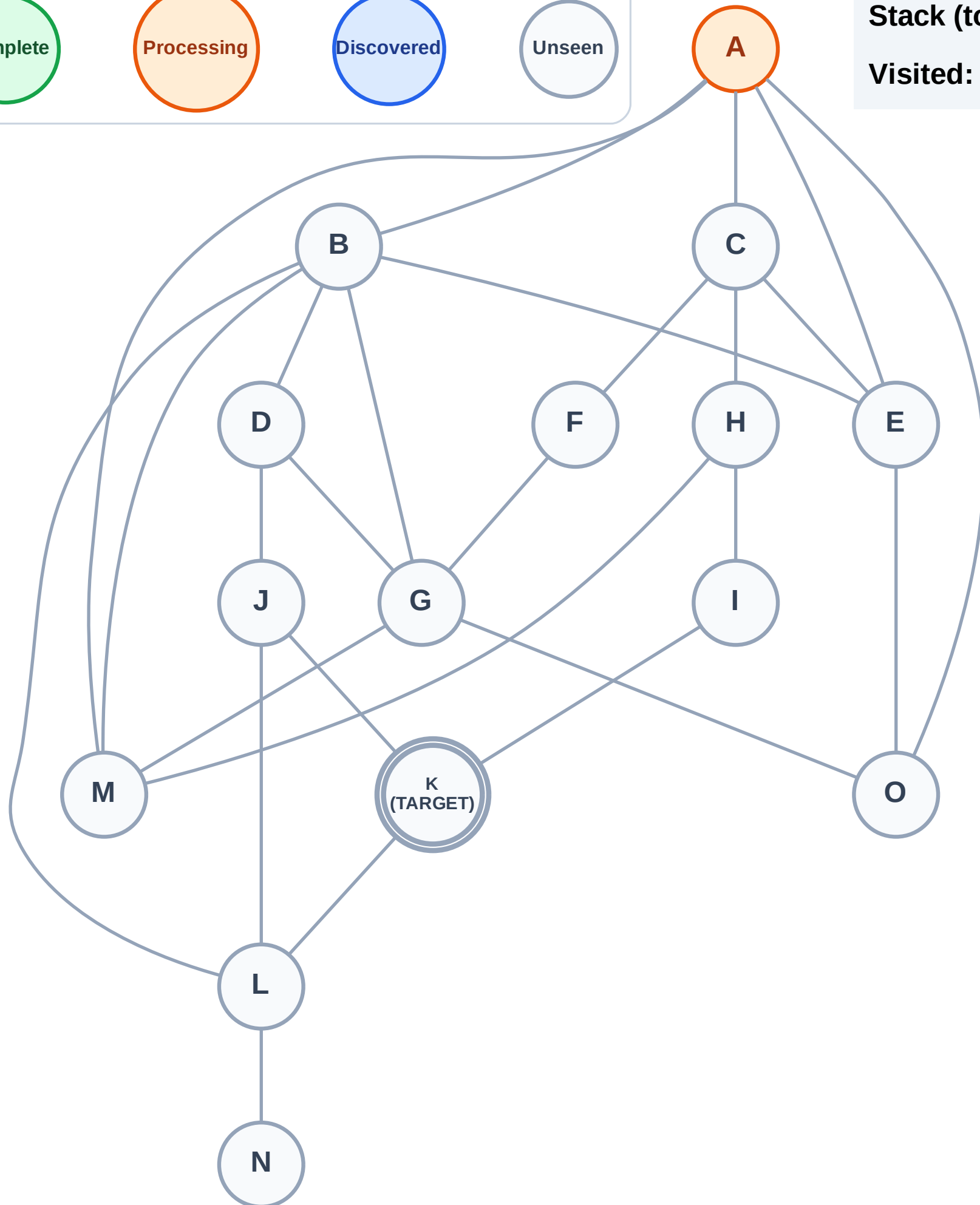
Step 2: Process node A

Legend



Stack (top at right): (empty)

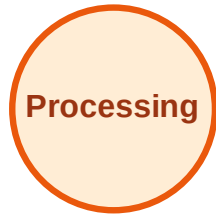
Visited: (none)



DFS Traversal

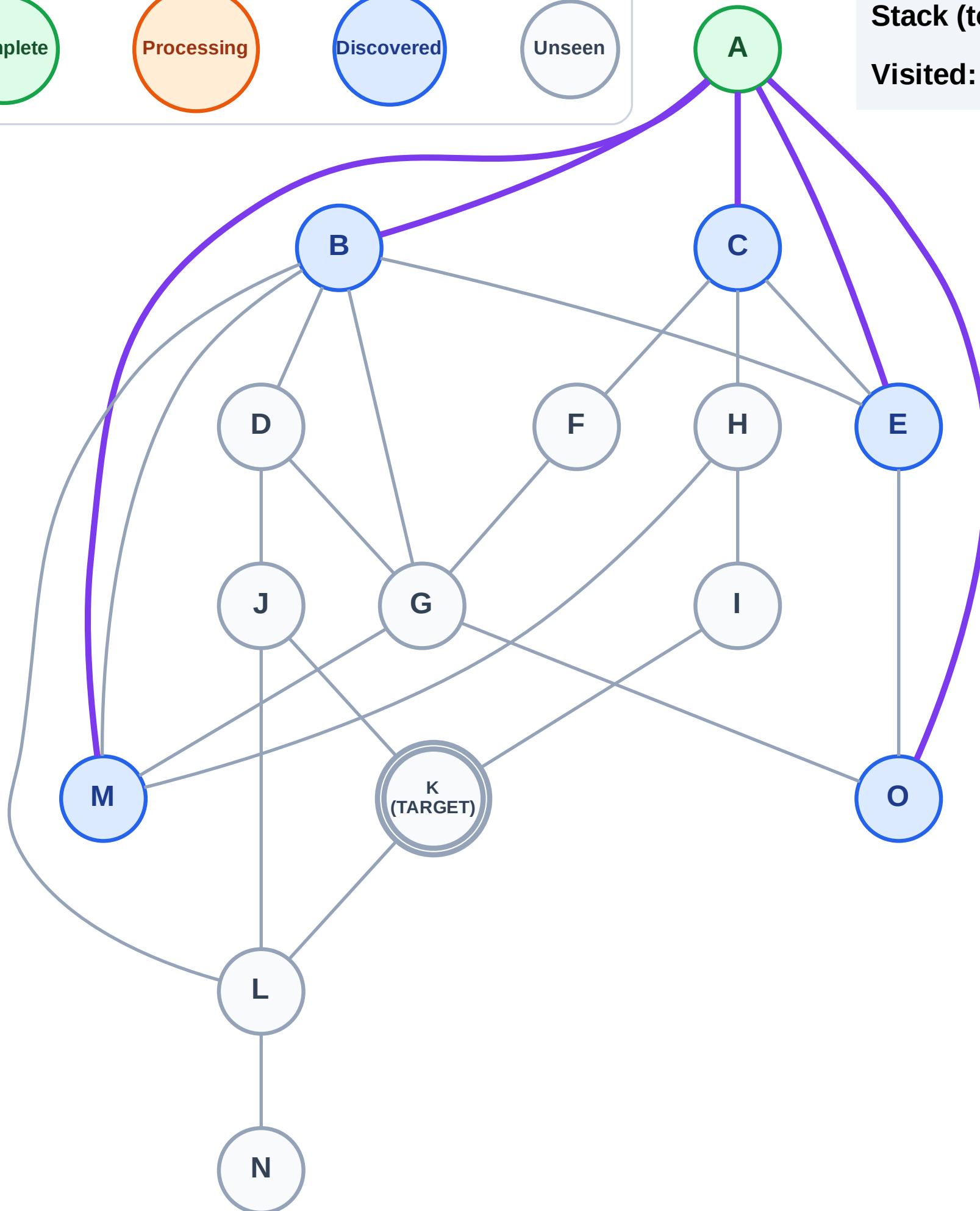
Step 3: Discover O, M, E, C, B from A

Legend



Stack (top at right): O | M | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

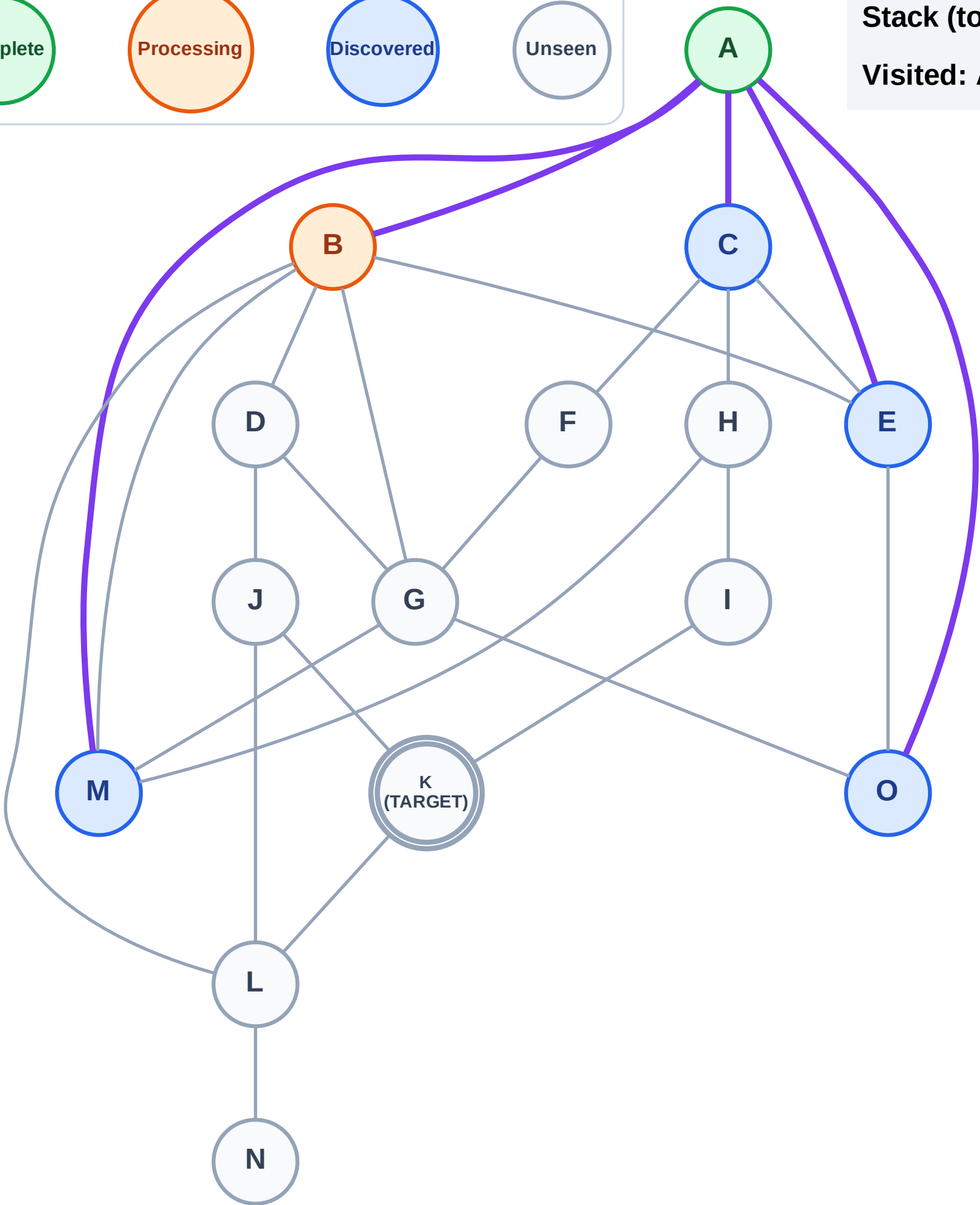
Processing

Discovered

Unseen

Stack (top at right): O | M | E | C

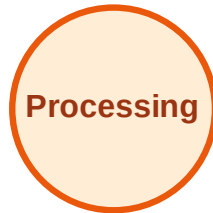
Visited: A



DFS Traversal

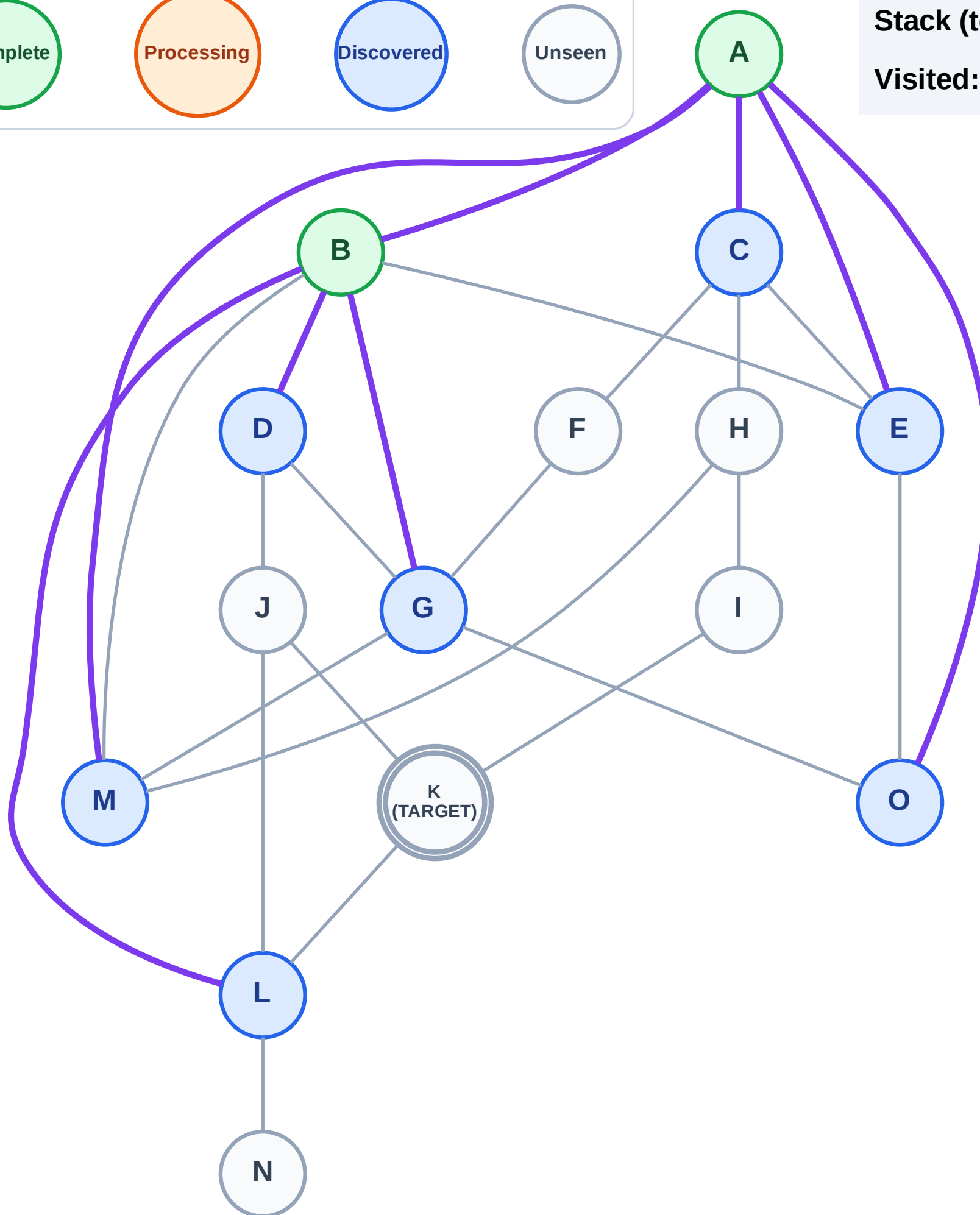
Step 5: Discover L, G, D from B

Legend



Stack (top at right): O | M | E | C | L | G | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

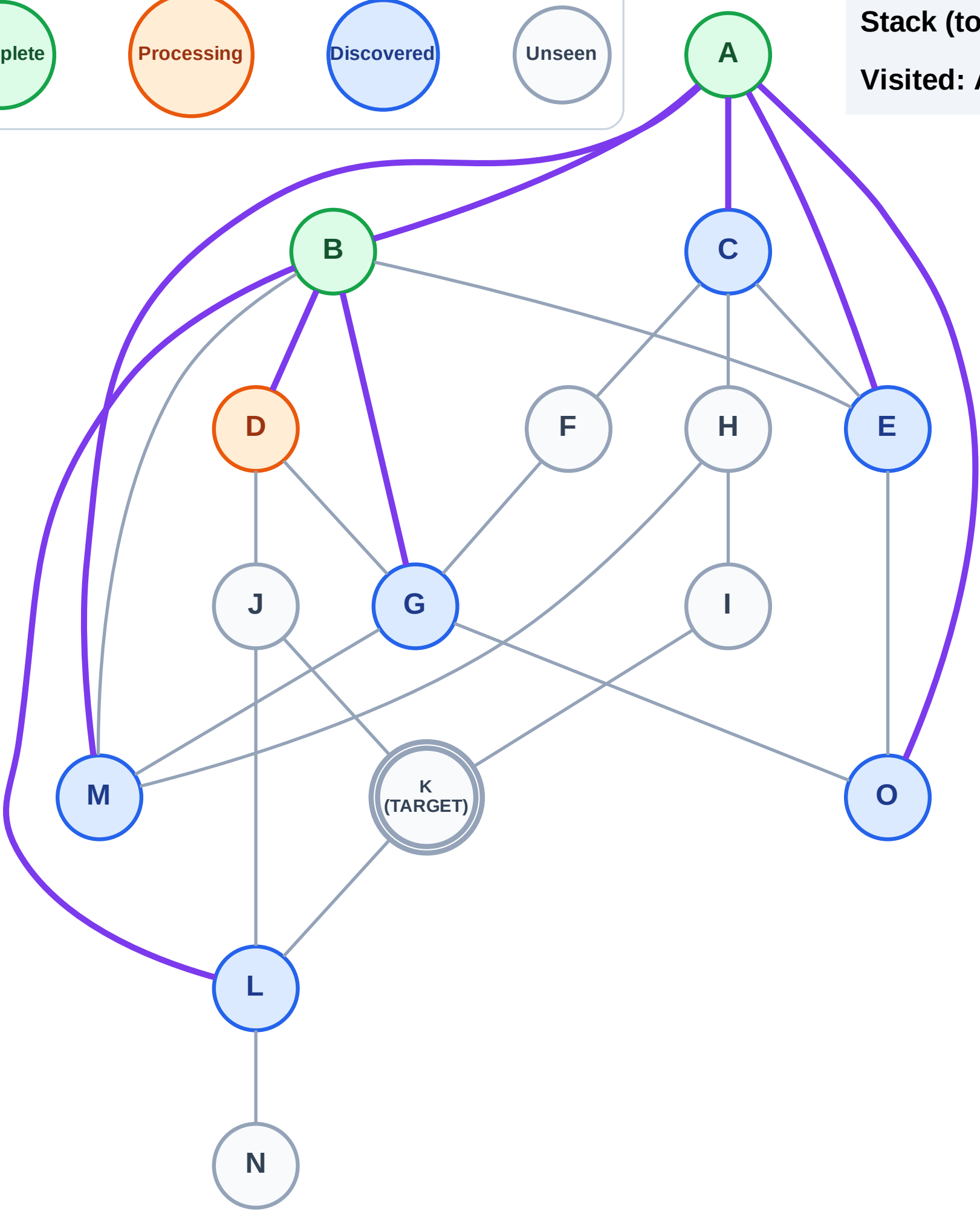
Processing

Discovered

Unseen

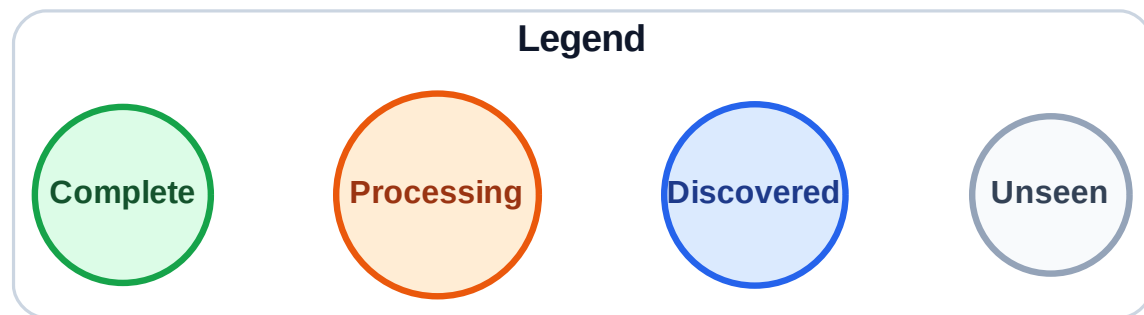
Stack (top at right): O | M | E | C | L | G

Visited: A, B

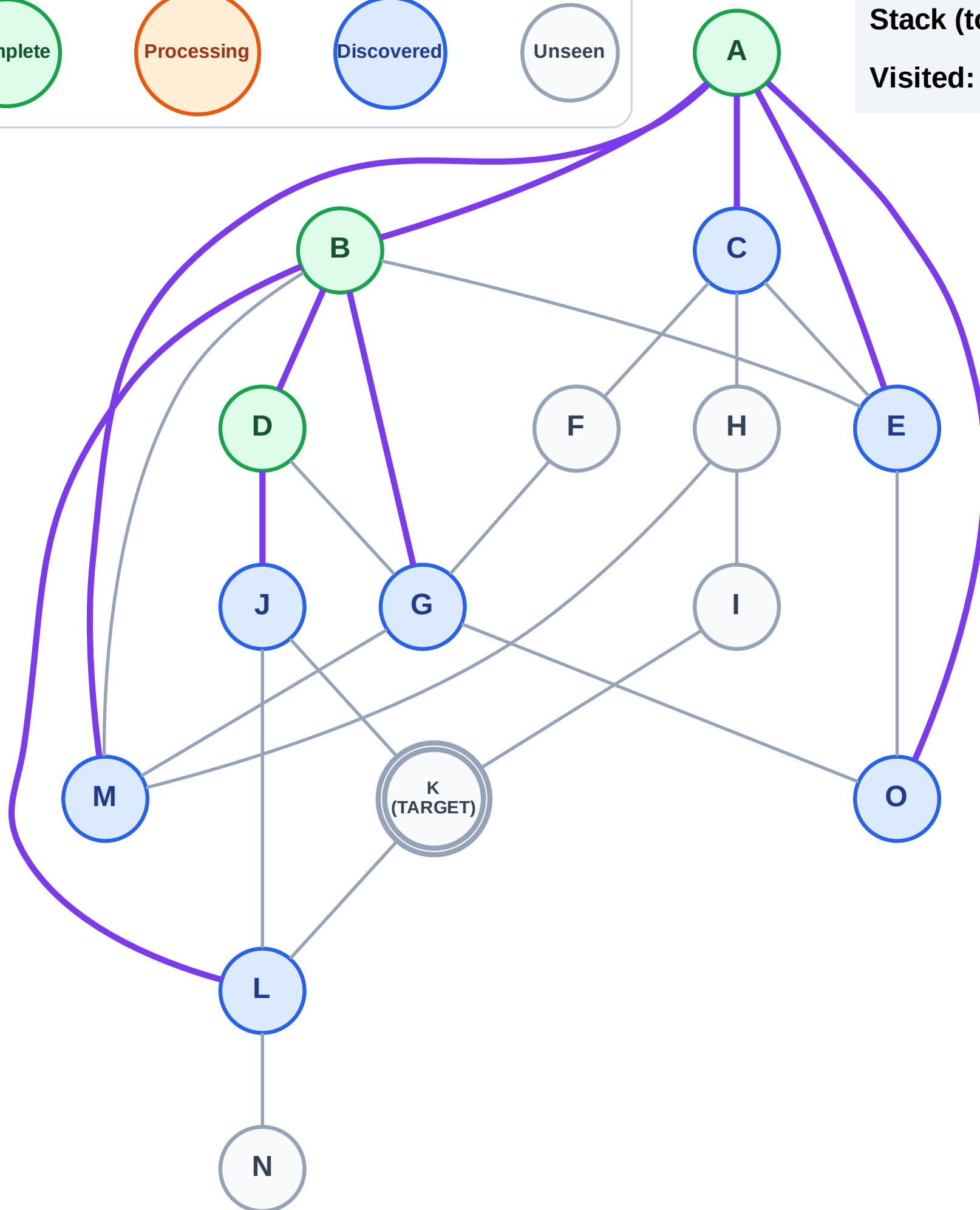


DFS Traversal

Step 7: Discover J from D



Stack (top at right): O | M | E | C | L | G | J
Visited: A, B, D



DFS Traversal

Step 8: Process node J

Legend

Complete

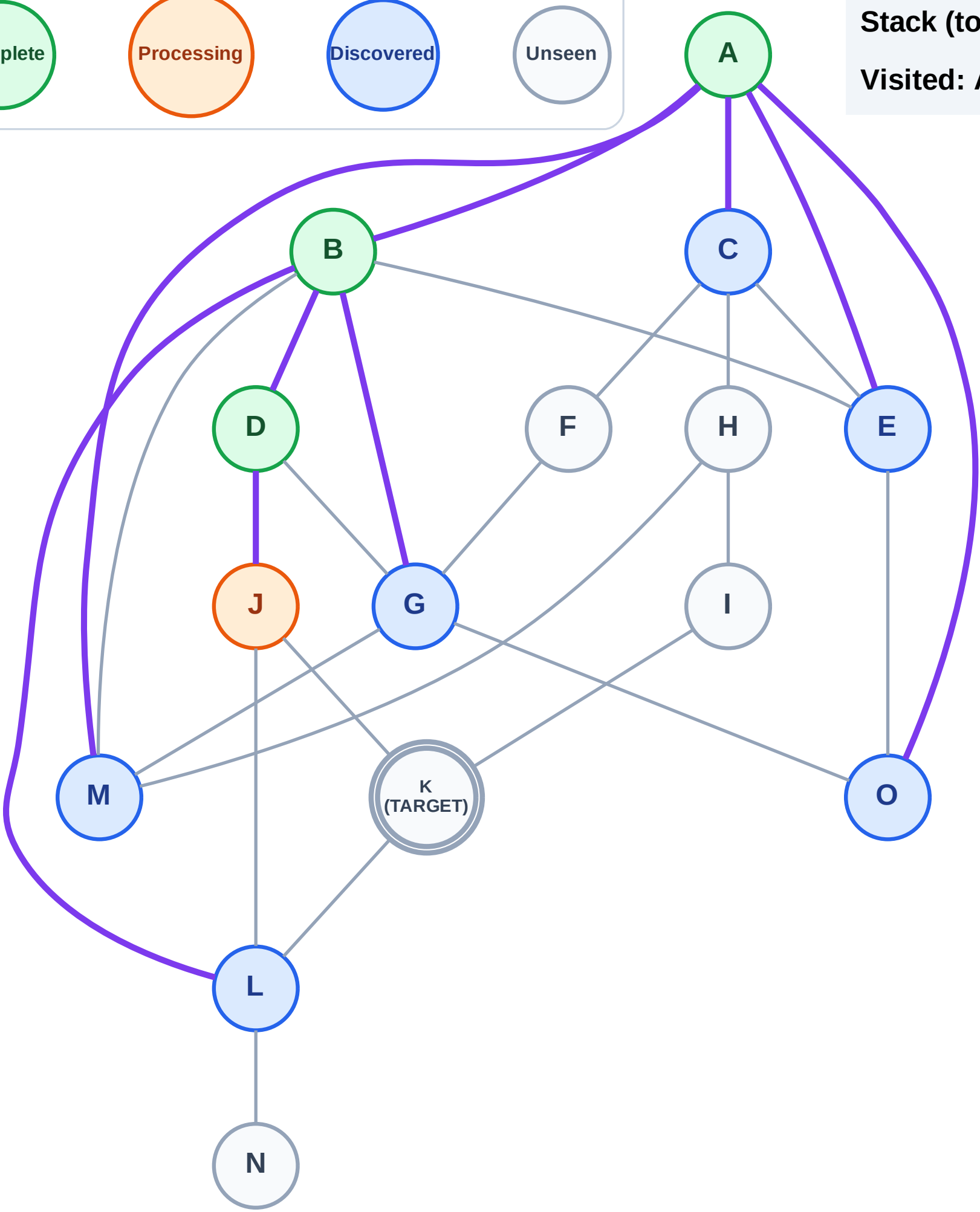
Processing

Discovered

Unseen

Stack (top at right): O | M | E | C | L | G

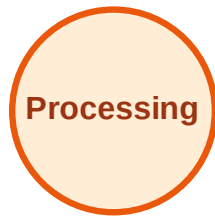
Visited: A, B, D



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → D → J → K

Legend



Stack (top at right): O | M | E | C | L | G

Visited: A, B, D, J, K

